

Mathematical Formulation of the Proactive Golden-Hour Bridge Model (PGBM) for Post-Earthquake UAV-Assisted Response

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Abstract—This document presents the mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM).

I. SURVIVOR PRIORITY AND SERVICE VALUE

For each detected survivor i , let $\sigma_i \in [0, 1]$ denote severity, $c_i \in [0, 1]$ denote detection confidence, and t_i^{det} denote the detection time. The time-dependent urgency of survivor i at decision time t is defined as

$$u_i(t) = \sigma_i \exp \left[-\alpha (t - t_i^{det}) \right], \quad \alpha = \frac{\ln 2}{T_{half}}, \quad (1)$$

where T_{half} is the urgency half-life. Thus, urgency decreases exponentially as the survivor waits for assistance.

To jointly account for survivor urgency and detection reliability, the effective service value is defined as

$$V_i(t) = \frac{2u_i(t)c_i}{u_i(t) + c_i}, \quad V_i(t) \in [0, 1]. \quad (2)$$

The harmonic-mean formulation ensures that a survivor receives a high service value only when both urgency and detection confidence are sufficiently high. A low value of either component reduces $V_i(t)$.

Let

$$x_i = \begin{cases} 1, & \text{if survivor } i \text{ is served,} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

The primary assignment objective is therefore to maximize the total effective service value of the survivors served by the available Responder UAVs:

$$\max \sum_{i \in S(t)} V_i(t)x_i. \quad (4)$$

The objective is subject to UAV payload, inventory, battery, and 3D reachability constraints. Hence, energy is treated as a feasibility constraint rather than an independent optimization objective. Confidence thresholding may additionally be applied as

$$c_i \geq c_{\text{thresh}}, \quad \forall i \text{ assigned to a tour.} \quad (5)$$